



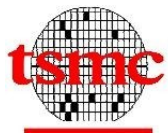
Taiwan Semiconductor Manufacturing Company Ltd.

No.8, Li-Hisn Rd.6, Science-Based Industrial Park, Hsin-Chu, Taiwan 300 R.O.C.

Submittal Comments Summary

Design

廠商名稱：漢唐集成股份有限公司 (Co. Ltd.) United Integrated Services	送審時間： (Date) 2025.01.24	TSMC DC No.: F22P2-S-220-030-2	
送審系統合約名稱 (Package)：F22P2 MEP & PX System - MEP Mechanical Sub-			
ject：Equipment Schedule table-1			
TSMC 彙總審核意見 (TSMC comments)：			
正本： To: 副本： C.C.:			
戳章 (Stamp)：			
Design Engineer	Construction Engineer	Construction Sec. Mgr.	Construction Project Mgr.



Taiwan Semiconductor Manufacturing Company Ltd.

25, Li-Hsin Rd., Science-Based Industrial Park, Hsin-Chu, Taiwan 300 R.O.C.

Tel : 886-3-563-6688 Ext.708-3210 Fax : 886-3-578-5890

Submittal Comments Summary

Design

廠商名稱：漢唐集成股份有限公司 (Co. Ltd.)	送審時間：2024/12/17 (Date)	TSMC DC No.: F22P2-S-220-030-1
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送審系統名稱 (Content): Equipment Schedule table -1

TSMC 彙總審核意見 (TSMC comments):

?/機械工程二課 盧勁甫(NO EXCEPTIONS NOTED) 2024/12/19 10:23

?/水處理工程一課 楊蹟翰(NO EXCEPTIONS NOTED) 2024/12/20 08:28

?/機械管理課 張皓浦(NO EXCEPTIONS NOTED) 2024/12/20 13:47

?/水處理工程一課 王弘柚(NO EXCEPTIONS NOTED) 2024/12/20 14:18

?/機械管理課 陳正賢(NO EXCEPTIONS NOTED) 2024/12/20 16:40

?/機械工程一課 黃柏勛(NO EXCEPTIONS NOTED) 2024/12/23 14:35

正本：漢唐集成股份有限公司

副本：

To:

C.C.:

戳章 (Stamp):

☒ NO EXCEPTIONS NOTED

☐ MAKE CORRECTIONS NOTED

☐ REVISE AND RESUBMIT

☐ SUBMIT ITEMS AS NOTED

Checking is only for general conformance with the design concept of the project and general compliance with the information given in the contract documents. Any action shown is subject to the requirements of the plans and specifications. Contractor is responsible for dimensions which shall be confirmed and correlated at the job site, fabrication processes and techniques of construction, coordination of the work with that of all other trades and the satisfactory performance of the work,



Project Team of Design & Construction Department

tsmc F22P2 MEP PROJECT SUBMITTAL RECOMMENT

F22P2-S-220-030-2 Equipment Schedule table-1

TSMC 審核意見	UIS 回覆
1.	1. 依業主要求，重新送審計畫。

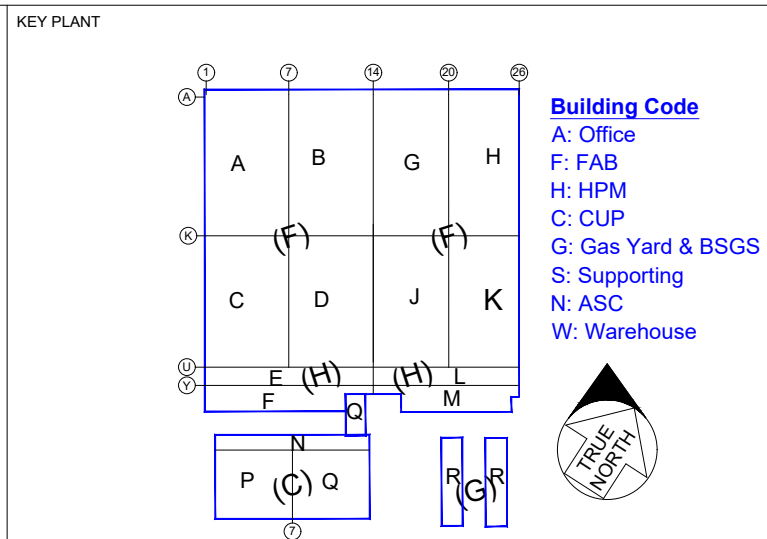
TSMC Submittal Plan				
Project Name:	Fab22 Phase-2	Date:	2024/8/20	
Package Name:	MEP & PX System - MEP Mechanical	Vendor:	漢唐集成股份有限公司	
Design				
Subject	Expected Date	Submittal Attachment		
		CD/Floppy	Drawing	Sample
送審計畫	2024/9/25			
時程計畫書(廠商重要里程碑計畫、廠商細部時程計畫、廠商人力需求計畫、廠商設備材料採購時程計畫、廠商下包商發包時程計畫、廠商重要設備及材料製造及運送時程計畫、廠商人員組織及職掌計畫、廠商施工策略及管理計畫、其他相關時程計畫...)	2024/10/16			
計算書				
計算書-Calculation sheet-Pipe Duct Support Seismic Calculation - CUP	2024/12/8			
計算書-Calculation sheet-Pipe Duct Support Seismic Calculation - FAB HPM	2024/12/8			
計算書-Calculation sheet-Pipe Duct pressure drop	2025/1/31			
計算書-Calculation sheet-Pipe Cooling load	2025/1/31			
計算書-Calcuation sheet-Pipe Pipe pressure drop	2025/1/31			
計算書-Calcuation sheet-各區域進氣量及換氣量	2025/3/26			
材料設備供應商送審資料 (Benchmark with exist FAB by 廠牌, 製造國家...)				
材料設備-DIW buffer Tank	2024/11/1			
材料設備-Pump (Chilled Water Condenser Water Warm Water RCW DI...etc.)	2024/10/19			
材料設備-Heat Exchanger	2024/10/19			
材料設備-Instrument (Pressure Temperatuer gauge...etc.)	2024/11/1			
材料設備-AHU PAHU FCU Air curtain...etc.	2024/11/27			
材料設備-PAK SF (Booster fan) for GEX reuse	2024/11/1			
材料設備-FAN (CO2 De-smoke Ventilation...etc.)	2024/11/1			
材料設備-Smoke Curtain	2024/11/29			
材料設備-Sand Filter (Tank pump pipe fitting...etc.)	2024/11/27			
材料設備-Bladder type expansion tank(Chilled Water Warm Water)	2024/10/30			
材料設備-Flexible Pipe connector Expansion Joint	2024/11/1			
材料設備-NG System (Shut off Valve Seamless Pipe...etc.)	2024/11/27			
材料設備-Valve (Butterfly Ball Gate Check Y-strainer)	2024/10/26			
材料設備-Chemical water treatment (Tank pump pipe fitting...etc.)	2024/11/29			
材料設備-Pipe (Galv. SUS PVC...etc)	2024/10/16			
材料設備-MAU	2024/11/3			
材料設備-Damper (Manual Fire Check Automatic...etc)	2024/11/27			
材料設備-Cooling coil Heating coil	2024/11/28			
材料設備-Insulation	2024/11/29			
材料設備-防火填塞	2024/11/29			
材料設備-高架地板	2024/11/29			
Drawings				
Equipment Schedule table -1	2024/12/8			

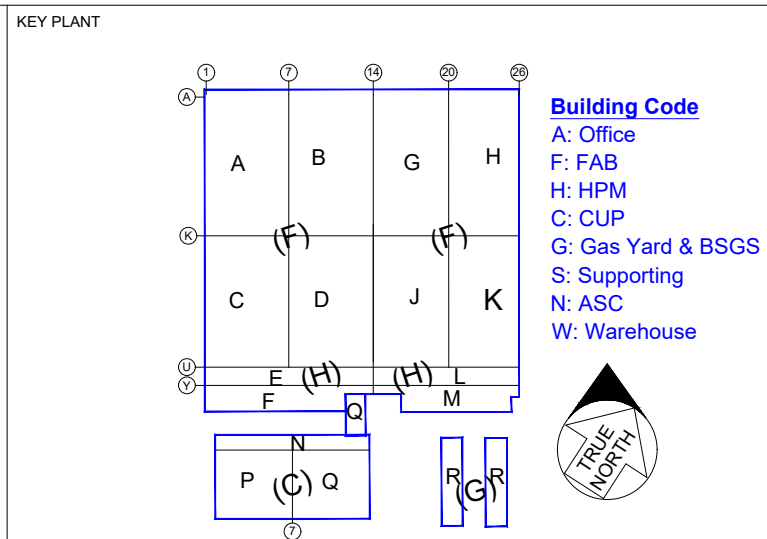
材料/設備 送審規格差異比較表

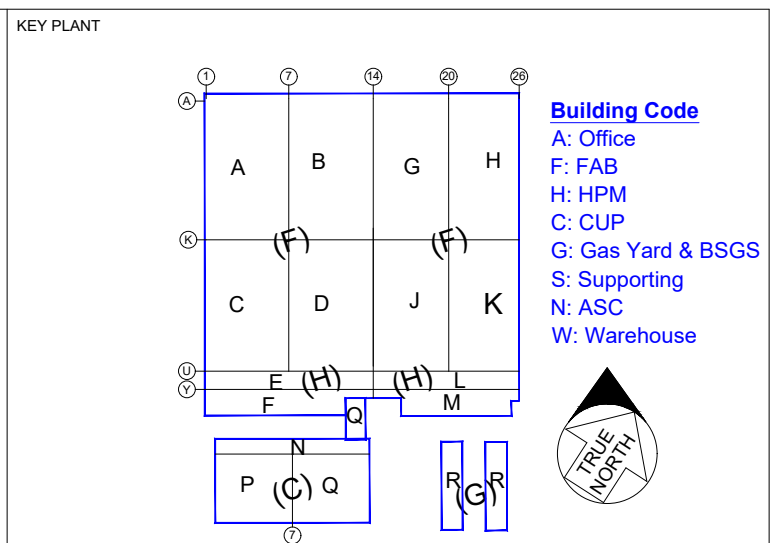
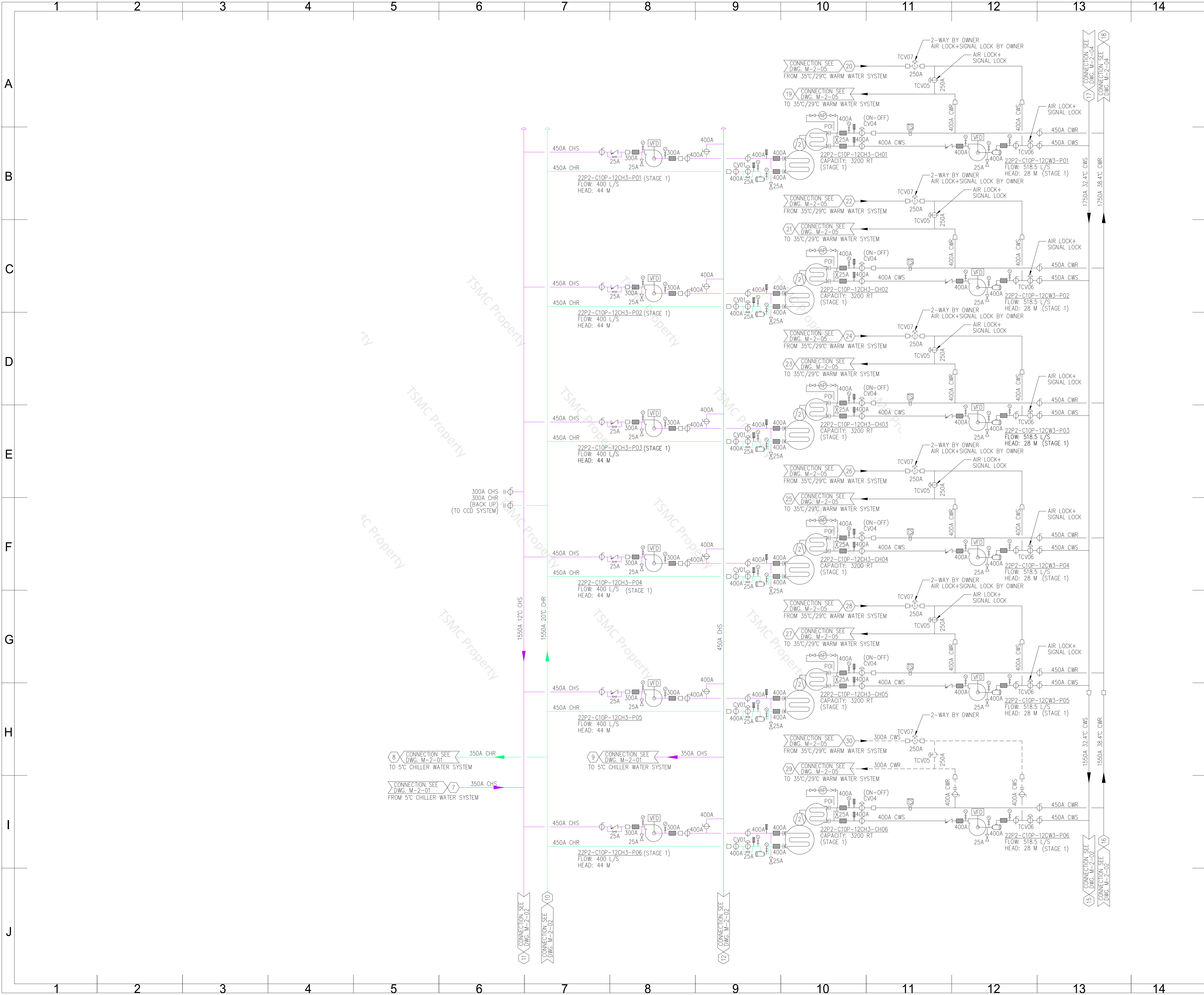
材料/設備 名稱 (請註明使用地點、設備、系統)	RFP			F20P1			F22P1			F22P2			是否符合 RFP	是否與 前廠相同	備註/差異說明 (如廠牌型號與前廠使用不同請詳細說明更換之原因)
	廠牌	型號	規格	廠牌	型號	規格	廠牌	型號	規格	廠牌	型號	規格			
Equipment Schedule table-1	NA	NA	NA	uis	uis	uis	uis	uis	同F20P1設計	uis	uis	同F22P1設計	是	是	

	1	2	3	4	5	6	7	8	9	10	11	12	13	14											
A	CENTRIFUGAL LIQUID CHILLER SCHEDULE																								
	UNIT MARK	DESIGN DUTY TONS	EVAPORATOR					CONDENSER					HEAT RECOVERY		HEAT RECOVERY		POWER		REMARKS						
B	22P2-C10P-5CH3-CH01	2600	L/S	EWT °C	LWT °C	PASS	WTR ΔP KPa	FOULING FACTOR	L/S	EWT °C	LWT °C	PASS	WTR ΔP KPa	FOULING FACTOR	HEAT RECOVERY KW	L/S	EWT °C	LWT °C	PASS	WTR ΔP KPa	FOULING FACTOR	CONSUMED POWER(KW) (A/B)	VOLTAGE	BY OWNER (STAGE 1) (STANDBY)	(TRANE)
	22P2-C10Q-5CH3-CH02	2600	272	13	5	1	16	0.0176	434.5	32.4	38.4	1	17.1	0.044	-	-	-	-	-	-	-	844.8/830.5	3/4160/60	BY OWNER (STAGE 1)	(TRANE)
	22P2-C10Q-5CH3-CH03	2600	272	13	5	1	16	0.0176	434.5	32.4	38.4	1	17.1	0.044	-	-	-	-	-	-	-	844.8/830.5	3/4160/60	BY OWNER (STAGE 1)	(TRANE)
	22P2-C10Q-5CH3-CH04	2600	272	13	5	1	16	0.0176	434.5	32.4	38.4	1	17.1	0.044	-	-	-	-	-	-	-	844.8/830.5	3/4160/60	BY OWNER (STAGE 1)	(TRANE)
	22P2-C10Q-5CH3-CH05	2600	272	13	5	1	16	0.0176	434.5	32.4	38.4	1	17.1	0.044	-	-	-	-	-	-	-	844.8/830.5	3/4160/60	BY OWNER (STAGE 1)	(TRANE)
	22P2-C10Q-5CH3-CH06	2600	272	13	5	1	16	0.0176	434.5	32.4	38.4	1	17.1	0.044	-	-	-	-	-	-	-	844.8/830.5	3/4160/60	BY OWNER (STAGE 1)	(TRANE)
	22P2-C10Q-5CH3-CH07	2600	272	13	5	1	16	0.0176	434.5	32.4	38.4	1	17.1	0.044	-	-	-	-	-	-	-	844.8/830.5	3/4160/60	BY OWNER (STAGE 1)	(TRANE)
	22P2-C10Q-5CH3-CH08	2600	272	13	5	1	16	0.0176	434.5	32.4	38.4	1	17.1	0.044	-	-	-	-	-	-	-	844.8/830.5	3/4160/60	BY OWNER (STAGE 1)	(TRANE)
	22P2-C10Q-5CH3-CH10	2600	272	13	5	1	16	0.0176	434.5	32.4	38.4	1	17.1	0.044	-	-	-	-	-	-	-	844.8/830.5	3/4160/60	BY OWNER (FUTURE)	(TRANE)
	22P2-C10Q-5CH3-CH11	2600	272	13	5	1	16	0.0176	434.5	32.4	38.4	1	17.1	0.044	-	-	-	-	-	-	-	844.8/830.5	3/4160/60	BY OWNER (FUTURE)	(TRANE)
C	22P2-C10Q-5CH3-CH12	2600	272	13	5	1	16	0.0176	434.5	32.4	38.4	1	17.1	0.044	-	-	-	-	-	-	-	844.8/830.5	3/4160/60	BY OWNER (FUTURE , SPACE)	(TRANE)
	22P2-C10P-12CH3-CH01	3200	335.7	20	12	1	23.6	0.0176	518.5	32.4	38.4	1	23.5	0.044	-	-	-	-	-	-	-	815.8/799.7	3/4160/60	BY OWNER (STAGE 1)	(TRANE)
	22P2-C10P-12CH3-CH02	3200	335.7	20	12	1	23.6	0.0176	518.5	32.4	38.4	1	23.5	0.044	-	-	-	-	-	-	-	815.8/799.7	3/4160/60	BY OWNER (STAGE 1)	(TRANE)
	22P2-C10P-12CH3-CH03	3200	335.7	20	12	1	23.6	0.0176	518.5	32.4	38.4	1	23.5	0.044	-	-	-	-	-	-	-	815.8/799.7	3/4160/60	BY OWNER (STAGE 1)	(TRANE)
	22P2-C10P-12CH3-CH04	3200	335.7	20	12	1	23.6	0.0176	518.5	32.4	38.4	1	23.5	0.044	-	-	-	-	-	-	-	815.8/799.7	3/4160/60	BY OWNER (STAGE 1)	(TRANE)
	22P2-C10P-12CH3-CH05	3200	335.7	20	12	1	23.6	0.0176	518.5	32.4	38.4	1	23.5	0.044	-	-	-	-	-	-	-	815.8/799.7	3/4160/60	BY OWNER (STAGE 1)	(TRANE)
	22P2-C10P-12CH3-CH06	3200	335.7	20	12	1	23.6	0.0176	518.5	32.4	38.4	1	23.5	0.044	-	-	-	-	-	-	-	815.8/799.7	3/4160/60	BY OWNER (STAGE 1)	(TRANE)
	22P2-C10Q-12CH3-CH07	3200	335.7	20	12	1	23.6	0.0176	518.5	32.4	38.4	1	23.5	0.044	-	-	-	-	-	-	-	815.8/799.7	3/4160/60	BY OWNER (STAGE 1)	(TRANE)
	22P2-C10Q-12CH3-CH08	3200	335.7	20	12	1	23.6	0.0176	518.5	32.4	38.4	1	23.5	0.044	-	-	-	-	-	-	-	815.8/799.7	3/4160/60	BY OWNER (STAGE 1)	(TRANE)
	22P2-C10Q-12CH3-CH09	3200	335.7	20	12	1	23.6	0.0176	518.5	32.4	38.4	1	23.5	0.044	-	-	-	-	-	-	-	815.8/799.7	3/4160/60	BY OWNER (STAGE 1)	(TRANE)
D	22P2-C10Q-12CH3-CH10	3200	335.7	20	12	1	23.6	0.0176	518.5	32.4	38.4	1	23.5	0.044	-	-	-	-	-	-	-	815.8/799.7	3/4160/60	BY OWNER (STAGE 1)	(TRANE)
	22P2-C10Q-12CH3-CH11	3200	335.7	20	12	1	23.6	0.0176	518.5	32.4	38.4	1	23.5	0.044	-	-	-	-	-	-	-	815.8/799.7	3/4160/60	BY OWNER (STAGE 1)	(TRANE)
	22P2-C10Q-12CH3-CH12	3200	335.7	20	12	1	23.6	0.0176	518.5	32.4	38.4	1	23.5	0.044	-	-	-	-	-	-	-	815.8/799.7	3/4160/60	BY OWNER (STAGE 1)	(TRANE)
	22P2-C10Q-12CH3-CH13	3200	335.7	20	12	1	23.6	0.0176	518.5	32.4	38.4	1	23.5	0.044	-	-	-	-	-	-	-	815.8/799.7	3/4160/60	BY OWNER (STAGE 1)	(TRANE)
	22P2-C10Q-12CH3-CH14	3200	335.7	20	12	1	23.6	0.0176	518.5	32.4	38.4	1	23.5	0.044	-	-	-	-	-	-	-	815.8/799.7	3/4160/60	BY OWNER (STAGE 1)	(TRANE)
	22P2-C10Q-12CH3-CH15	3200	335.7	20	12	1	23.6	0.0176	518.5	32.4	38.4	1	23.5	0.044	-	-	-	-	-	-	-	815.8/799.7	3/4160/60	BY OWNER (FUTURE)	(TRANE)
	22P2-C10Q-12CH3-CH16	3200	335.7	20	12	1	23.6	0.0176	518.5	32.4	38.4	1	23.5	0.044	-	-	-	-	-	-	-	815.8/799.7	3/4160/60	BY OWNER (FUTURE)	(TRANE)
	22P2-C10Q-12CH3-CH17	3200	335.7	20	12	1	23.6	0.0176	518.5	32.4	38.4	1	23.5	0.044	-	-	-	-	-	-	-	815.8/799.7	3/4160/60	BY OWNER (FUTURE , SPACE)	(TRANE)
	E	COOLING TOWER CELL SCHEDULE																							
		UNIT MARK	L/S	EWT °C	LWT °C	AIR TEMP °C WB	MOTOR			TYPE	REMARKS														
KW		VOLTAGE	TYPE																						
22P2-CR2P-CDW3-CT01		1017	38.4	32.4	29.4	190	3/460/60	TEFC	COUNTERFLOW	W/VFD , (STAGE 1) BY OWNER (STANDBY)															
22P2-CR2P-CDW3-CT02		1017	38.4	32.4	29.4	190	3/460/60	TEFC	COUNTERFLOW	W/VFD , (STAGE 1) BY OWNER															
22P2-CR2P-CDW3-CT03		1017	38.4	32.4	29.4	190	3/460/60	TEFC	COUNTERFLOW	W/VFD , (STAGE 1) BY OWNER															
22P2-CR2P-CDW3-CT04		1017	38.4	32.4	29.4	190	3/460/60	TEFC	COUNTERFLOW	W/VFD , (STAGE 1) BY OWNER															
22P2-CR2P-CDW3-CT05		1017	38.4	32.4	29.4	190	3/460/60	TEFC	COUNTERFLOW	W/VFD , (STAGE 1) BY OWNER															
22P2-CR2P-CDW3-CT06		1017	38.4	32.4	29.4	190	3/460/60	TEFC	COUNTERFLOW	W/VFD , (STAGE 1) BY OWNER															
22P2-CR2Q-CDW3-CT07		1017	38.4	32.4	29.4	190	3/460/60	TEFC	COUNTERFLOW	W/VFD , (STAGE 1) BY OWNER															
22P2-CR2Q-CDW3-CT08		1017	38.4	32.4	29.4	190	3/460/60	TEFC	COUNTERFLOW	W/VFD , (STAGE 1) BY OWNER															
22P2-CR2Q-CDW3-CT09		1017	38.4	32.4	29.4	190	3/460/60	TEFC	COUNTERFLOW	W/VFD , (STAGE 1) BY OWNER															
22P2-CR2Q-CDW3-CT10		1017	38.4	32.4	29.4	190	3/460/60	TEFC	COUNTERFLOW	W/VFD , (STAGE 1) BY OWNER															
22P2-CR2Q-CDW3-CT11		1017	38.4	32.4	29.4	190	3/460/60	TEFC	COUNTERFLOW	W/VFD , (STAGE 1) BY OWNER															
22P2-CR2Q-CDW3-CT12	1017	38.4	32.4	29.4	190	3/460/60	TEFC	COUNTERFLOW	W/VFD , (STAGE 1) BY OWNER																
22P2-CR2Q-CDW3-CT13	1017	38.4	32.4	29.4	190	3/460/60	TEFC	COUNTERFLOW	W/VFD , (FUTURE) BY OWNER																
22P2-CR2Q-CDW3-CT14	1017	38.4	32.4	29.4	190	3/460/60	TEFC	COUNTERFLOW	W/VFD , (FUTURE) BY OWNER																
F	PLATE TYPE HEAT EXCHANGER SCHEDULE																								
	UNIT MARK	SERVICE	CAPACITY KW	COLD SIDE				FLUID	HOT SIDE				FLUID	REMARKS											
	EWT °C	LWT °C	L/S	WTR ΔP KPa	EWT °C	LWT °C	L/S		WTR ΔP KPa																
	22P2-C10P-35WW3-HX01	35°C WARM WATER SYSTEM	8200	29	35	327	<50	WATER	38.4	32.4	327	<50	WATER	BY OWNER (STANDBY) (STAGE 1)	(SWEP)										
	22P2-C10P-35WW3-HX02	35°C WARM WATER SYSTEM	8200	29	35	327	<50	WATER	38.4	32.4	327	<50	WATER	BY OWNER (STAGE 1)	(SWEP)										
	22P2-C10P-35WW3-HX03	35°C WARM WATER SYSTEM	8200	29	35	327	<50	WATER	38.4	32.4	327	<50	WATER	BY OWNER (STAGE 1)	(SWEP)										
	22P2-C10P-35WW3-HX04	35°C WARM WATER SYSTEM	8200	29	35	327	<50	WATER	38.4	32.4	327	<50	WATER	BY OWNER (STAGE 1)	(SWEP)										
22P2-C10P-35WW3-HX05	35°C WARM WATER SYSTEM	8200	29	35	327	<50	WATER	38.4	32.4	327	<50	WATER	BY OWNER (STAGE 1)	(SWEP)											
22P2-C10P-35WW3-HX06	35°C WARM WATER SYSTEM	8200	29	35	327	<50	WATER	38.4	32.4	327	<50	WATER	BY OWNER (FUTURE)	(SWEP)											
G																									
H																									
I																									
J																									
	1	2	3	4	5	6	7	8	9	10	11	12	13	14											

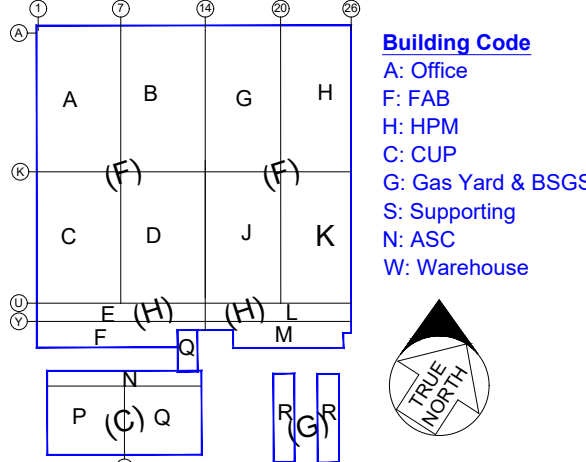
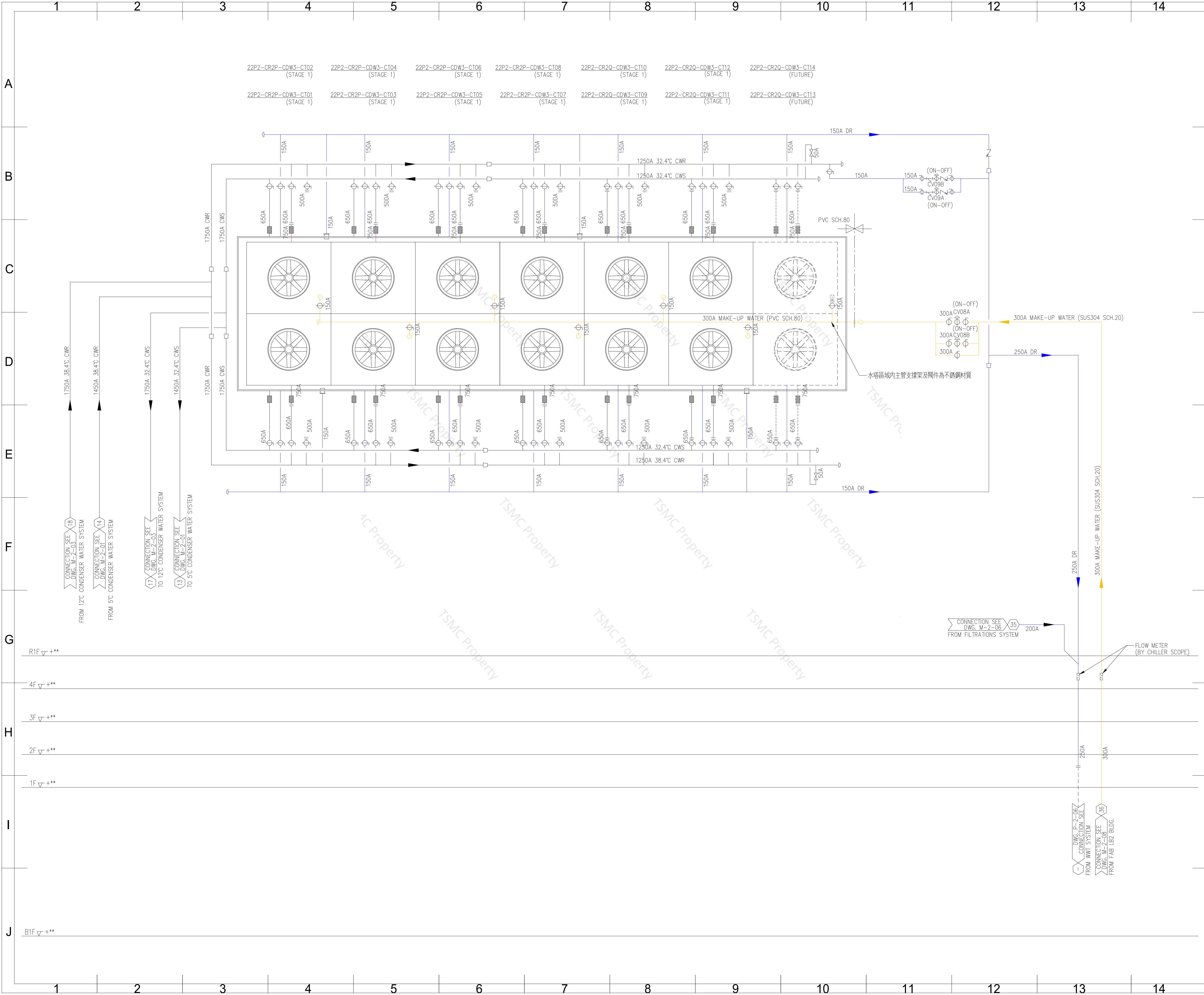
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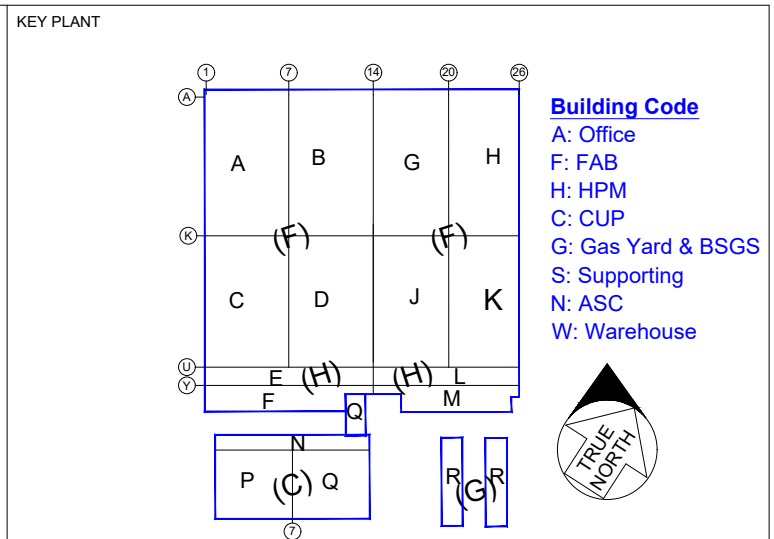
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台積電 TSMC			
台灣積體電路製造股份有限公司 Taiwan Semiconductor Manufacturing Company, Ltd.			
ACCEPTANCE REVIEWED	DATE	ACCEPTANCE REVIEWED	DATE
CONTRACTOR iis 漢唐集成股份有限公司 United Integrated Service			
DRAWN	TCM	DATE	2024/12/3
DESIGNED	TCM	DATE	2024/12/3
TITLE FLOW DIAGRAM 12°C/20°C CHILLED & CONDENSER WATER SYSTEM			
PROJECT FAB22 Phase 2 PROJECT			
PACKAGE NAME MEP SYSTEM			
DRAWING NAME F22P2-M-2-C-03			
SCALE	UNIT		
NOT TO SCALE	mm		
FILE NAME	Rev		
F22P2-M-2-C-03			



核准	日期	修正者	修正
VER.	DATE	REV BY	REVISION
台灣積體電路製造股份有限公司 Taiwan Semiconductor Manufacturing Company, Ltd.			
ACCEPTANCE REVIEWED	DATE	ACCEPTANCE REVIEWED	DATE
CONTRACTOR 漢唐集成股份有限公司 United Integrated Service			
DRAWN	TCM	DATE	2024/12/3
DESIGNED	TCM	DATE	2024/12/3
TITLE FLOW DIAGRAM CHILLED & CONDENSER WATER SYSTEM			
PROJECT FAB22 Phase 2 PROJECT			
PACKAGE NAME MEP SYSTEM			
DRAWING NAME F22P2-M-2-C-04			
SCALE	NOT TO SCALE		
FILE NAME	F22P2-M-2-C-04		
UNIT	mm		
Rev			



Name: X:\G200002_Drawings\MEP\CUP\M\F22P2-M-2-C-07.dwg User: nicky-3121 Date: 31/11/24 - 9:47am